

Paper Review:

Functional linear non-Gaussian acyclic model for causal discovery

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· Yang, T. L., Lee, K. Y., Zhang, K., & Suzuki, J. (2024).
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Behaviormetrika, 51(2), 567-588.

Main contributions and outline

Main contributions

- Extend the LiNGAM to the **functional** linear non-Gaussian acyclic model (Func-LiNGAM).
- Establish **identifiability** for the Func-LiNGAM.
- Propose an **algorithm** to learn the Func-LiNGAM based on the functional principal component analysis (FPCA) regression.

- Preliminaries
- LiNGAM for random vectors
- LiNGAM for random functions
- Algorithm
- Empirical results
- Conclusion

Preliminaries

Proposition 1 (Darmois-Skitovich theorem).

Define two random variables L_1 and L_2 as linear combinations of independent random variables ξ_j ($j = 1, \dots, m$) for $m \geq 2$, i.e.,

$$L_1 = \sum_{j=1}^m \alpha_j \xi_j, \quad L_2 = \sum_{j=1}^m \beta_j \xi_j.$$

Then, if L_1 and L_2 are independent, all random variables ξ_j with $\alpha_j \beta_j \neq 0$ are Gaussian.

Proposition 2 (Identifiability for the LiNGAM).

Assume that X follows the LiNGAM. Then, a random variable X_j is exogenous if and only if X_j is independent of its residuals $r_k(j) = X_k - \frac{\text{Cov}(X_k, X_j)}{\text{Var}(X_j)} X_j$ for all $j \neq k$.

Hilbert space

Definition 1 (Banach space).

A *Banach space* is a **complete normed vector space**.

Definition 2 (Hilbert space).

A *Hilbert space* is a Banach space equipped with an **inner product** that induces the norm.

Proposition 3 (Cardinality of a Hilbert space).

A Hilbert space is *separable* if and only if it has a **countable** orthonormal basis.

- In the sequel, **any Hilbert space is endowed with separability**.

Linear operator

Definition 3 (Linear operator).

Suppose $\mathbb{H}_1$ and $\mathbb{H}_2$ are two Hilbert spaces. A mapping $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$ is called a *linear operator* if

$$T(af + bg) = aT(f) + bT(g) \quad \text{for all scalars } a, b \text{ and } f, g \in \mathbb{H}_1.$$

In addition, T is said to be *bounded* if there exists a positive constant C such that

$$\|T(f)\|_{\mathbb{H}_2} \leq C\|f\|_{\mathbb{H}_1}.$$

Definition 4 (Continuity of a linear operator).

A linear operator T is said to be *continuous* if $T(f_n) \rightarrow T(f)$ whenever $f_n \rightarrow f$.

Proposition 4.

A linear operator T is **bounded if and only if it is continuous**.

Linear operator

Proposition 5 (Adjoint operator).

Let $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$ be a bounded linear operator. Then, there exists a **unique** bounded linear operator $T^* : \mathbb{H}_2 \rightarrow \mathbb{H}_1$ such that

- $\langle T(f), g \rangle_{\mathbb{H}_2} = \langle f, T^*(g) \rangle_{\mathbb{H}_1},$
- $\|T^*\|_{op} = \|T\|_{op},$ and
- $(T^*)^* = T.$

The linear operator T^* is called the *adjoint* of T .

Definition 5 (Symmetric operator).

A bounded linear operator $T : \mathbb{H} \rightarrow \mathbb{H}$ is said to be *symmetric* if $T = T^*$.

Linear operator

Definition 6 (Compact operator).

Let $\mathbb{H}_1$ and $\mathbb{H}_2$ be two Hilbert spaces and B be a closed unit ball in $\mathbb{H}_1$. A linear operator $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$ is called a *compact* operator if $T(B)$ has a compact closure.

Proposition 6.

A linear operator $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$ is compact if and only if for any bounded sequence $\{f_n\}_{n=1}^{\infty}$ in $\mathbb{H}_1$, $\{T(f_n)\}_{n=1}^{\infty}$ has a convergent subsequence in $\mathbb{H}_2$.

Proposition 7.

Let $T : \mathbb{H} \rightarrow \mathbb{H}$ be a linear operator. Then, the followings hold:

- If T is compact, then T is bounded.
- T is compact if and only if T^* is compact.
- If T is compact, then TT^* and T^*T are also compact.

Proposition 8 (Spectral theorem).

Suppose that $T : \mathbb{H} \rightarrow \mathbb{H}$ is a **compact symmetric** operator. Then, there exist an **orthonormal basis** $\{\varphi_k\}_{k=1}^{\infty}$ of $\mathbb{H}$ and $\{\lambda_k\}_{k=1}^{\infty}$ in $\mathbb{R}$ such that

$$T(\varphi_k) = \lambda_k \varphi_k, \quad \forall k \geq 1.$$

Moreover, for any $\epsilon > 0$ there are only **finitely many** λ_k with $|\lambda_k| > \epsilon$. Conversely, every linear operator of the above form is compact and symmetric.

Here, each λ_k is called an *eigenvalue* of T , and each φ_k an *eigenvector*.

- When $\dim \mathbb{H} = \infty$, one can also deduce that $\lambda_k \rightarrow 0$ as $k \rightarrow \infty$.
- This implies that **any compact symmetric operator cannot be invertible in infinite dimensional cases**.

Random function

Definition 7 (Random function).

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(\mathbb{H}, \mathcal{B}(\mathbb{H}))$ be a measurable space, where $\mathbb{H}$ is a Hilbert space. Then, a measurable mapping $X : \Omega \rightarrow \mathbb{H}$ is called a *random function*.

Definition 8 (Expectation of a random function).

Let $X : \Omega \rightarrow \mathbb{H}$ be a random function. Under the condition that the expectation of $\|X\|_{\mathbb{H}}$, $\int_{\Omega} \|X\|_{\mathbb{H}} dP$, is bounded, the *mean* of X is defined by the Bochner integral as $\int_{\Omega} X dP$.

Definition 9 (Covariance operator of a random function).

Let $X : \Omega \rightarrow \mathbb{H}$ be a random function such that $E(\|X\|_{\mathbb{H}}^2) < \infty$. A linear operator $C : \mathbb{H} \rightarrow \mathbb{H}$ defined by

$$\langle C(x), y \rangle_{\mathbb{H}} = E\left(\langle X - E(X), x \rangle_{\mathbb{H}} \cdot \langle X - E(X), y \rangle_{\mathbb{H}}\right)$$

is called a *covariance operator*. It is known that C is **compact and symmetric**.

Gaussian random function

Definition 10 (Gaussian random function).

A random function X taking values in $\mathbb{H}$ is called a *Gaussian random function* with mean μ and covariance operator C , if $\langle X, y \rangle_{\mathbb{H}}$ is Gaussian for all $y \in \mathbb{H}$, i.e.,

$$\langle X, y \rangle_{\mathbb{H}} \sim N\left(\langle \mu, y \rangle_{\mathbb{H}}, \langle C(y), y \rangle_{\mathbb{H}}\right), \quad \forall y \in \mathbb{H}.$$

Definition 11 (Jointly Gaussian random function).

Random functions X_j taking values in $\mathbb{H}_j$ for $j = 1, \dots, k$ are *jointly Gaussian* if $\langle X_1, y_1 \rangle_{\mathbb{H}_1}, \dots, \langle X_k, y_k \rangle_{\mathbb{H}_k}$ are jointly Gaussian for all $y_j \in \mathbb{H}_j$ for $j = 1, \dots, k$.

Proposition 9.

Suppose that two random functions X_1 and X_2 are jointly Gaussian. Then, X_1 and X_2 are **independent if and only if they are uncorrelated**.

LiNGAM for random vectors

Proposition 10 (Multivariate Darmois-Skitovich theorem).

For $m \geq 2$, let L_1 and L_2 be defined by

$$L_1 = \sum_{j=1}^m A_j \xi_j, \quad L_2 = \sum_{j=1}^m B_j \xi_j,$$

where ξ_j are independent random vectors and A_j, B_j are matrices. Then, if L_1 and L_2 are independent, all random vectors ξ_j with **invertible** matrices A_j and B_j are Gaussian.

Theorem 1 (Identifiability of the LiNGAM for random vectors).

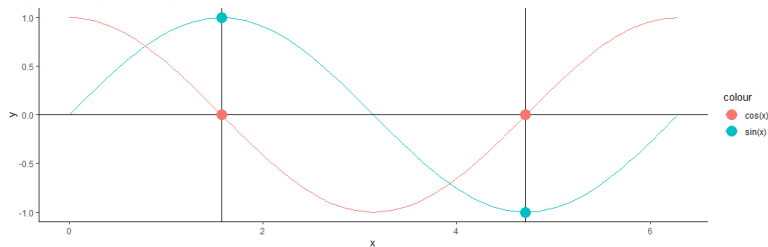
Let

$$X_1 = \epsilon_1, \quad X_2 = A_{12}X_1 + \epsilon_2,$$

where ϵ_1 and ϵ_2 are independent random vectors and A_{12} is an **invertible** matrix. If ϵ_1 or ϵ_2 is non-Gaussian, then the model is identifiable.

LiNGAM for random functions

Motivating example



- Consider a simple example:
 - $\mathbb{H} = L^2[0, 2\pi]$.
 - $X_1 = \sin(\cdot) + \epsilon_1$ and $X_2 = T_{12}(X_1) + \epsilon_2$, where $T_{12} : \mathbb{H} \rightarrow \mathbb{H}$ is a linear operator such that $T_{12}(f) = f((\cdot + \pi/2) \bmod 2\pi)$.
- If we have realizations for each random function as $\mathbf{x}_1^{(1)} = (1, -1)$ and $\mathbf{x}_2^{(1)} = (0, 0)$, a naive application of the LiNGAM for random vectors would yield a wrong result with $\hat{T}_{12} = 0$.

Proposition 11 (Functional Darmois-Skitovich theorem).

For $m \geq 2$, let L_1 and L_2 be defined by

$$L_1 = \sum_{j=1}^m A_j(\xi_j), \quad L_2 = \sum_{j=1}^m B_j(\xi_j),$$

where ξ_j are independent random functions and A_j, B_j are **bounded** operators. Then, if L_1 and L_2 are independent, all random functions ξ_j with **invertible** operators A_j and B_j are Gaussian.

Theorem 2 (Identifiability of the LiNGAM for random functions).

Let

$$X_1 = \epsilon_1, \quad X_2 = T_{12}(X_1) + \epsilon_2,$$

where ϵ_1 and ϵ_2 are independent random functions and T_{12} is an **invertible bounded** operator. If ϵ_1 or ϵ_2 is non-Gaussian, then the model is identifiable.

LiNGAM for random functions in multivariate scenarios

Proposition 2 (Identifiability for the LiNGAM).

Assume that X follows the LiNGAM. Then, a random variable X_j is exogenous if and only if X_j is independent of its residuals $r_k(j) = X_k - \frac{\text{Cov}(X_k, X_j)}{\text{Var}(X_j)} X_j$ for all $j \neq k$.

Lemma 1.

Assume that X follows the Func-LiNGAM. Then, a random function X_j is exogenous if and only if X_j is independent of its residuals $r_k(j) = X_k - T_{jk} X_j$ for all $j \neq k$.

Conjecture 1.

Assume that X follows the Func-LiNGAM. In addition, assume that $\dim \mathbb{H}_j < \infty$ for all $j = 1, \dots, p$. Then, a random function X_j is exogenous if and only if X_j is independent of its residuals $r_k(j) = X_k - C_{kj} C_{jj}^{-1} X_j$ for all $j \neq k$, where

- $\langle C_{kj}(x), y \rangle_{\mathbb{H}_k} = E\left(\langle X_j - E(X_j), x \rangle_{\mathbb{H}_j} \cdot \langle X_k - E(X_k), y \rangle_{\mathbb{H}_k}\right),$
- C_{jj} is the covariance operator of X_j , and C_{jj}^{-1} is its inverse.

Pairwise regression when $\dim \mathbb{H}_j = \infty$

- Suppose we solve a pairwise regression problem in the Func-LiNGAM:

$$X_k = T_{j \rightarrow k} X_j + \tilde{\epsilon}_k,$$

where j precedes k .

- Assuming that the eigenvalues of C_{jj} are **all positive**, $T_{j \rightarrow k}$ is identifiable.

($\because$) Let $\{\lambda_\ell\}_{\ell=1}^\infty$ and $\{\varphi_{j,\ell}\}_{\ell=1}^\infty$ be the eigenvalues and eigenvectors of C_{jj} , respectively. Since $\{\varphi_{j,\ell}\}_{\ell=1}^\infty$ is the orthonormal basis of $\mathbb{H}_j$, we have

$$C_{jj}f = 0 \iff \sum_{\ell=1}^{\infty} \lambda_\ell \langle f, \varphi_{j,\ell} \rangle_{\mathbb{H}_j} \varphi_{j,\ell} = 0 \iff \|f\|_{\mathbb{H}_j} = 0 \iff f = 0 \quad (C_{jj} \text{ is injective}).$$

Using this fact, one can show that $T_{j \rightarrow k} X_j = \tilde{T}_{j \rightarrow k} X_j$ a.s. implies $T_{j \rightarrow k} = \tilde{T}_{j \rightarrow k}$. □

Pairwise regression when $\dim \mathbb{H}_j = \infty$

- The **conventional OLS method** to estimate $T_{j \rightarrow k}$ is to solve

$$\arg \min_{T: \mathbb{H}_j \rightarrow \mathbb{H}_k, \text{bounded}} E \|X_k - TX_j\|_{\mathbb{H}_k}^2 = \arg \min_{T: \mathbb{H}_j \rightarrow \mathbb{H}_k, \text{bounded}} \text{Tr}(C_{kk}) - 2\text{Tr}(TC_{kj}) + \text{Tr}(TC_{jj}T^*).$$

- Taking the Fréchet derivative and setting to zero yields the **normal equation**

$$T_{\text{est}}C_{jj} - C_{kj} = 0 \iff T_{\text{est}}C_{jj} = C_{kj}.$$

- However, according to the spectral theorem, C_{jj} is **not invertible** when $\dim \mathbb{H}_j = \infty$, which implies that we cannot recover the true operator.

- Intuitively, this is because **the variance of X_j vanishes in several directions**:

$$\lambda_\ell = \text{Var}(\langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j}) \rightarrow 0 \text{ as } \ell \rightarrow \infty,$$

where $\{\lambda_\ell\}_{\ell=1}^\infty$ and $\{\varphi_{j,\ell}\}_{\ell=1}^\infty$ are the eigenvalues and eigenvectors of C_{jj} , respectively.

Pairwise regression when $\dim \mathbb{H}_j = \infty$

Proposition 11 (Functional Darmois-Skitovich theorem).

For $m \geq 2$, let L_1 and L_2 be defined by

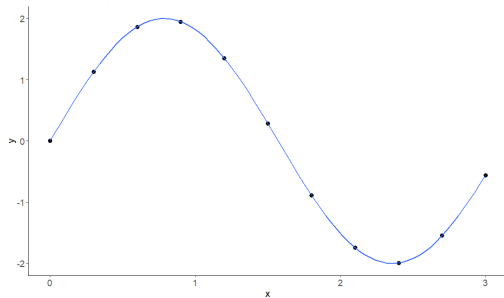
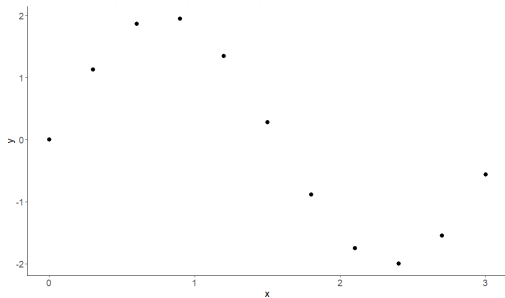
$$L_1 = \sum_{j=1}^m A_j(\xi_j), \quad L_2 = \sum_{j=1}^m B_j(\xi_j),$$

where ξ_j are independent random functions and A_j, B_j are **bounded** operators. Then, if L_1 and L_2 are independent, all random functions ξ_j with **invertible** operators A_j and B_j are Gaussian.

- Consequently, when $\dim \mathbb{H}_j = \infty$, the alternative solution $T_{est} = C_{kj}C_{jj}^{-}$ **fails to be invertible**, and thus Proposition 11 does not apply.
- Therefore, it is necessary to restrict $\mathbb{H}_1, \dots, \mathbb{H}_p$ to have **finite dimensions**.

Algorithm

Function recovery



- In real-world scenarios, it is likely to observe **incomplete realizations** of each sample function rather than its complete form.
- Various **interpolation** methods (e.g., spline, kernel) are applicable.

Proposition 12 (Kosambi–Karhunen–Loève theorem).

Let $X : \Omega \rightarrow \mathbb{H}$ be a random function such that $E(\|X\|_{\mathbb{H}}^2) < \infty$, and let C be the covariance operator of X . Then,

$$E\left\| (X - E(X)) - \sum_{k=1}^N \langle X - E(X), \varphi_k \rangle_{\mathbb{H}} \varphi_k \right\|_{\mathbb{H}}^2 \rightarrow 0 \text{ as } N \rightarrow \infty,$$

where $\{\varphi_k\}_{k=1}^{\infty}$ comprises the eigenvectors of C .

- Consider a pairwise regression problem in the Func-LiNGAM:

$$T_{est} \in \arg \min_{T: \mathbb{H}_j \rightarrow \mathbb{H}_k, \text{bounded}} E\|X_k - TX_j\|_{\mathbb{H}_k}^2.$$

- Let $\{\varphi_{j,\ell}\}_{\ell=1}^{\infty}$ and $\{\varphi_{k,\ell}\}_{\ell=1}^{\infty}$ be the eigenvectors of C_{jj} and C_{kk} , respectively. According to Proposition 12 and the continuity of T , we have

$$E\|X_k - TX_j\|_{\mathbb{H}_k}^2 = E\left\| \sum_{\ell=1}^{\infty} \langle X_k, \varphi_{k,\ell} \rangle_{\mathbb{H}_k} \varphi_{k,\ell} - \sum_{\ell=1}^{\infty} \langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j} T\varphi_{j,\ell} \right\|_{\mathbb{H}_k}^2.$$

FPCA regression

- Recall that

$$E\|X_k - TX_j\|_{\mathbb{H}_k}^2 = E\left\|\sum_{\ell=1}^{\infty} \langle X_k, \varphi_{k,\ell} \rangle_{\mathbb{H}_k} \varphi_{k,\ell} - \sum_{\ell=1}^{\infty} \langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j} T\varphi_{j,\ell}\right\|_{\mathbb{H}_k}^2.$$

- In addition, considering that $\{\varphi_{k,\ell}\}_{\ell=1}^{\infty}$ is an **orthonormal basis** for $\mathbb{H}_k$, we have

$$T\varphi_{j,\ell} = \sum_{\ell'=1}^{\infty} \beta_{\ell,\ell'} \varphi_{k,\ell'}$$

for some $\beta_{\ell,\ell'}$. In particular, $\beta_{\ell,\ell'} = \langle T\varphi_{j,\ell}, \varphi_{k,\ell'} \rangle_{\mathbb{H}_k}$.

- Then, the above equation can be written as

$$E\left\|\sum_{\ell=1}^{\infty} \langle X_k, \varphi_{k,\ell} \rangle_{\mathbb{H}_k} \varphi_{k,\ell} - \sum_{\ell=1}^{\infty} \langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j} T\varphi_{j,\ell}\right\|_{\mathbb{H}_k}^2 = E\left\|\sum_{\ell'=1}^{\infty} \left(\langle X_k, \varphi_{k,\ell'} \rangle_{\mathbb{H}_k} - \sum_{\ell=1}^{\infty} \beta_{\ell,\ell'} \langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j}\right) \varphi_{k,\ell'}\right\|_{\mathbb{H}_k}^2.$$

- Lastly, by the **Parseval identity**, we arrive at

$$E\left\|\sum_{\ell'=1}^{\infty} \left(\langle X_k, \varphi_{k,\ell'} \rangle_{\mathbb{H}_k} - \sum_{\ell=1}^{\infty} \beta_{\ell,\ell'} \langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j}\right) \varphi_{k,\ell'}\right\|_{\mathbb{H}_k}^2 = \sum_{\ell'=1}^{\infty} E\left(\langle X_k, \varphi_{k,\ell'} \rangle_{\mathbb{H}_k} - \sum_{\ell=1}^{\infty} \beta_{\ell,\ell'} \langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j}\right)^2.$$

FPCA regression

- In practice we **truncate** each series at M_j and M_k basis functions:

$$\sum_{\ell'=1}^{\infty} E\left(\langle X_k, \varphi_{k,\ell'} \rangle_{\mathbb{H}_k} - \sum_{\ell=1}^{\infty} \beta_{\ell,\ell'} \langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j}\right)^2 \approx \sum_{\ell'=1}^{M_k} E\left(\langle X_k, \varphi_{k,\ell'} \rangle_{\mathbb{H}_k} - \sum_{\ell=1}^{M_j} \beta_{\ell,\ell'} \langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j}\right)^2$$

- Now, since $\langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j}$ are **mutually uncorrelated**, the given regression problem boils down to $M_j \times M_k$ separate problems:

$$\tilde{\beta}_{\ell,\ell'} := \frac{\langle C_{kj} \varphi_{j,\ell}, \varphi_{k,\ell'} \rangle_{\mathbb{H}_k}}{\langle C_{jj} \varphi_{j,\ell}, \varphi_{j,\ell} \rangle_{\mathbb{H}_j}} = \arg \min_{\beta_{\ell,\ell'} \in \mathbb{R}} E\left(\langle X_k, \varphi_{k,\ell'} \rangle_{\mathbb{H}_k} - \beta_{\ell,\ell'} \langle X_j, \varphi_{j,\ell} \rangle_{\mathbb{H}_j}\right)^2, \quad \ell = 1, \dots, M_j, \quad \ell' = 1, \dots, M_k.$$

- Then, we construct $\tilde{T}$ as

$$\tilde{T}f = \sum_{\ell=1}^{M_j} \sum_{\ell'=1}^{M_k} \tilde{\beta}_{\ell,\ell'} \langle f, \varphi_{j,\ell} \rangle_{\mathbb{H}_j} \varphi_{k,\ell'}.$$

- If $\dim \mathbb{H}_j = M_j$ and $\dim \mathbb{H}_k = M_k$, $\tilde{T} = C_{kj} C_{jj}^{-1}$ holds.

Algorithm

- **Input:** $Wp \times n$ data matrix, and B-spline basis functions $\{\varphi_{j,\ell}\}_{\ell=1}^{M_j}$ for $j = 1, \dots, p$
 - Each p function has W realization points.
 - Typical B-spline bases are not orthonormal, and hence the Gram-Schmidt process should be preceded.
- **Output:** Adjacent matrix $\hat{T} \in \mathbb{R}^{p \times p}$

1. Step 1: Function recovery

- Recover $X_j^{(i)}$ by $\hat{X}_j^{(i)}(\cdot) = \mathfrak{B}_j(\cdot; X_j^{(i)}(t_1), \dots, X_j^{(i)}(t_W))$ for $i = 1, \dots, n, j = 1, \dots, p$.

2. Step 2: Dimension reduction

- Take the principal components of $\hat{X}_j^{(i)}$ by $\hat{X}_{j,\text{PC}}^{(i)} = (\langle \hat{X}_j^{(i)}, \varphi_{j,1} \rangle_{\mathfrak{B}_j}, \dots, \langle \hat{X}_j^{(i)}, \varphi_{j,M_j} \rangle_{\mathfrak{B}_j})$ for $i = 1, \dots, n, j = 1, \dots, p$.

3. Step 3: LiNGAM procedure

- Initialize an ordered list of variables $K = \emptyset$ and $m = 1$.
- Repeat until $p - 1$ subscripts are appended to K :
 - (a) For $j \in V \setminus K$, perform least squares regressions of $\hat{X}_{k,PC}$ on $\hat{X}_{j,PC}$ for all $k \in V \setminus K$ ($k \neq j$) and compute the residual vectors $r_k(j)$ as

$$[r_k(j)^{(i)}]_{\ell'} = [\hat{X}_{k,PC}^{(i)}]_{\ell'} - \hat{\beta}_{1,\ell'}[\hat{X}_{j,PC}^{(i)}]_1 - \cdots - \hat{\beta}_{M_j,\ell'}[\hat{X}_{j,PC}^{(i)}]_{M_j}, \quad \ell' = 1, \dots, M_k, i = 1, \dots, n,$$

where $\hat{\beta}_{\ell,\ell'}$ is the regression coefficient between $[\hat{X}_{k,PC}^{(i)}]_{\ell'}$ and $[\hat{X}_{j,PC}^{(i)}]_{\ell}$.

- (b) Find a subscript m that satisfies

$$m = \arg \min_{j \in V \setminus K} IM(\hat{X}_{j,PC}; \mathbf{r}(j)),$$

where IM is an independence measure such as mutual information or other measures.

- (c) Append m to the end of K .
 - (d) Update $\hat{X}_{k,PC}^{(i)} = r_k(m)^{(i)}$ for all $k \in V \setminus K$ and $i = 1, \dots, n$.

Empirical results

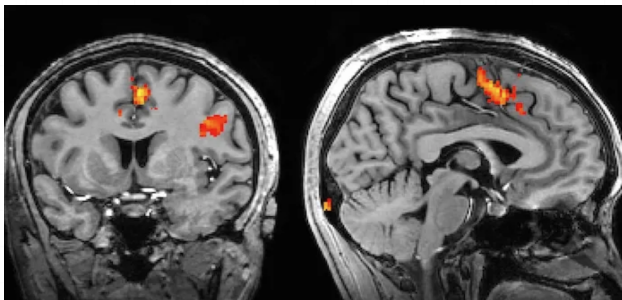
Experiment

- $X_j(\cdot) = X_{j-1}(\cdot) + \Phi(\cdot)^T \epsilon_j$, where $X_0(\cdot) \equiv 0$, $\Phi(x) = (e^{-4\pi ix}, e^{-2\pi ix}, 1, e^{2\pi ix}, e^{4\pi ix})$, and $\epsilon_j \sim \chi^2(I_5)$.
 - Note that $X_j \in \overline{\text{span}\{e^{-4\pi ix}, \dots, e^{4\pi ix}\}} \subset L^2[0, 1]$ for $j = 1, \dots, p$.
 - The true graph of the above model is $1 \rightarrow \dots \rightarrow p$.
- $n \in \{100, 200, 300, 700\}$ and $p \in \{5, 10, 20, 30, 50, 70\}$.
- Every random function is realized at $W = 1000$ equidistance points, $0 = t_1, \dots, t_{1000} = 1$.
- The observed random function is $Z_j(t_k) = X_j(t_k) + 0.25\chi^2(1)$.
- The number of B-spline bases are set at $M = 5$.
 - If the cubic B-spline is adopted, the number of knots is $1 + 2 + 1 = 4$.

Table 1 Evaluation of Func-LiNGAM with various number p of functions

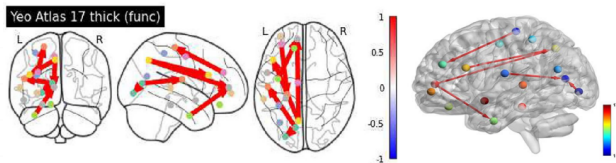
Data size	Metrics	Various number of functions (mean $\pm$ standard deviation)					
		$p = 5$	$p = 10$	$p = 20$	$p = 30$	$p = 50$	$p = 70$
$n = 100$	Precision	0.76 ± 0.14	0.64 ± 0.10	0.57 ± 0.09	0.40 ± 0.06	0.30 ± 0.04	0.25 ± 0.03
	Recall	0.99 ± 0.04	0.95 ± 0.0	0.90 ± 0.07	0.75 ± 0.07	0.65 ± 0.04	0.59 ± 0.04
	F1	0.85 ± 0.10	0.76 ± 0.08	0.70 ± 0.09	0.52 ± 0.07	0.41 ± 0.05	0.35 ± 0.03
	SHD	1.40 ± 0.95	5.03 ± 1.91	13.47 ± 4.17	33.47 ± 6.56	74.73 ± 9.86	119.47 ± 10.70
$n = 200$	Precision	0.83 ± 0.14	0.76 ± 0.29	0.72 ± 0.07	0.70 ± 0.06	0.54 ± 0.05	0.46 ± 0.07
	Recall	1.00 ± 0.00	0.80 ± 0.24	0.99 ± 0.01	0.97 ± 0.03	0.88 ± 0.03	0.81 ± 0.07
	F1	0.90 ± 0.08	0.78 ± 0.27	0.83 ± 0.05	0.81 ± 0.05	0.67 ± 0.05	0.59 ± 0.07
	SHD	0.97 ± 0.91	3.63 ± 4.58	7.70 ± 2.35	12.53 ± 3.36	37.03 ± 6.60	66.20 ± 12.79
$n = 300$	Precision	0.85 ± 0.13	0.79 ± 0.28	0.75 ± 0.07	0.74 ± 0.05	0.70 ± 0.05	0.60 ± 0.04
	Recall	1.00 ± 0.00	0.84 ± 0.23	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.93 ± 0.03
	F1	0.92 ± 0.08	0.81 ± 0.26	0.86 ± 0.05	0.85 ± 0.03	0.82 ± 0.03	0.73 ± 0.04
	SHD	0.80 ± 0.75	3.17 ± 4.43	6.57 ± 2.50	10.27 ± 2.41	21.27 ± 4.36	42.90 ± 6.25
$n = 700$	Precision	0.92 ± 0.10	0.81 ± 0.08	0.80 ± 0.07	0.78 ± 0.05	0.74 ± 0.03	0.70 ± 0.02
	Recall	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.95 ± 0.05
	F1	0.96 ± 0.06	0.88 ± 0.05	0.88 ± 0.04	0.87 ± 0.03	0.85 ± 0.02	0.83 ± 0.05
	SHD	0.40 ± 0.55	2.50 ± 1.20	4.96 ± 2.06	8.80 ± 2.34	17.40 ± 2.97	32.70 ± 4.37

Real data



- Functional MRI (fMRI) is a part of magnetic resonance imaging (MRI) that deals with the brain's functional activity in the rest and the active states.
- The data consist of 155 subjects ($n = 155$), 168 time points ($W = 168$), and 17 parcels ($p = 17$).
- The participants were instructed to watch a movie that aimed to evoke various brain activities.

Real data



- To check the non-Gaussianity assumption, the Shapiro-Wilk normality test was performed.
- Each parcel was represented by $M = 5$ principal components.

Conclusion

Conclusion

- This paper discuss **the identifiability of the Func-LiNGAM** based on the functional Darmois-Skitovich theorem.
- **All operators involved in the model must be invertible**, which constrains the func-LiNGAM to finite dimensional Hilbert spaces.
- The experimental results reveal that **the func-LiNGAM struggles with variable selection**.
- Yet **the consistency of the proposed algorithm remains unexplored**, it appears to hold provided that the underlying B-spline theory is well established.

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